02/10/2025 — Clase # 27 — Electrodinámica

$\vec{E}, \vec{D} \mid \vec{B}, \vec{H}$

Unit 2: Electric fields in dielectric media.

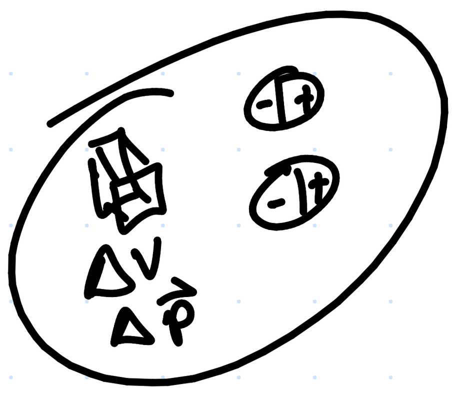
ε

χ

α

$\vec{D}$

$\vec{E}$



$$\vec{P} = \lim_{\Delta V \to 0} \frac{\Delta \vec{P}}{\Delta V} = \vec{P}(x,y,z)$$

t – tensor

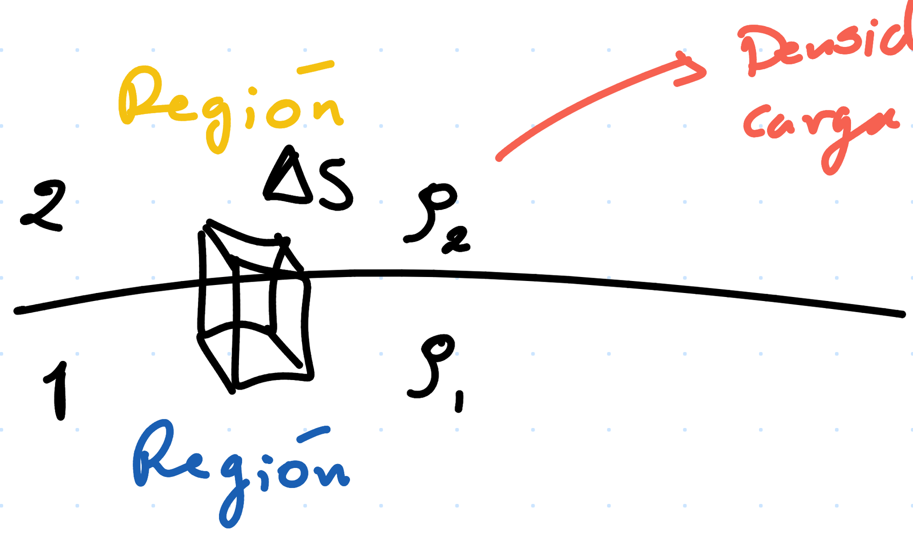
Anisotropy

$$\vec{P} = \chi \vec{E} \; ; \; \vec{D} = \epsilon \vec{E}$$

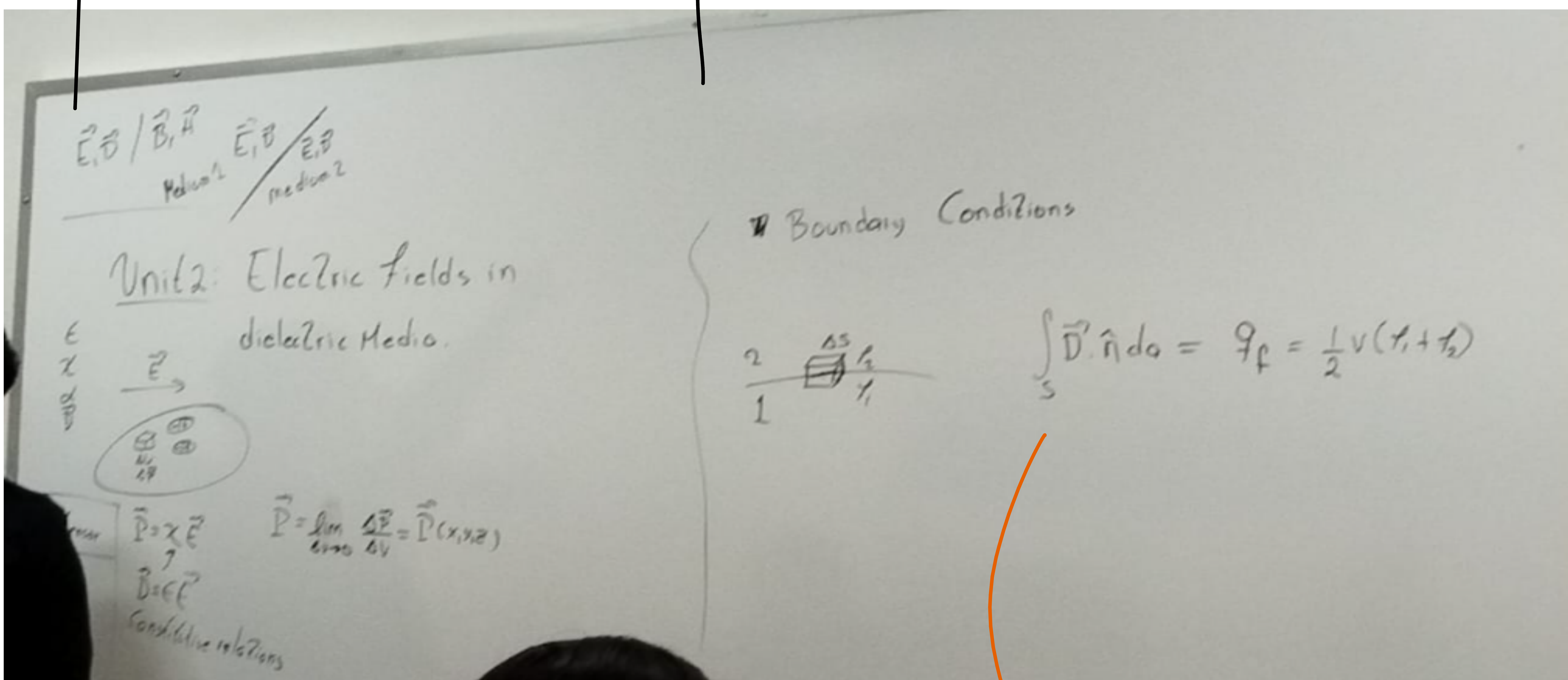
Constitutive relations

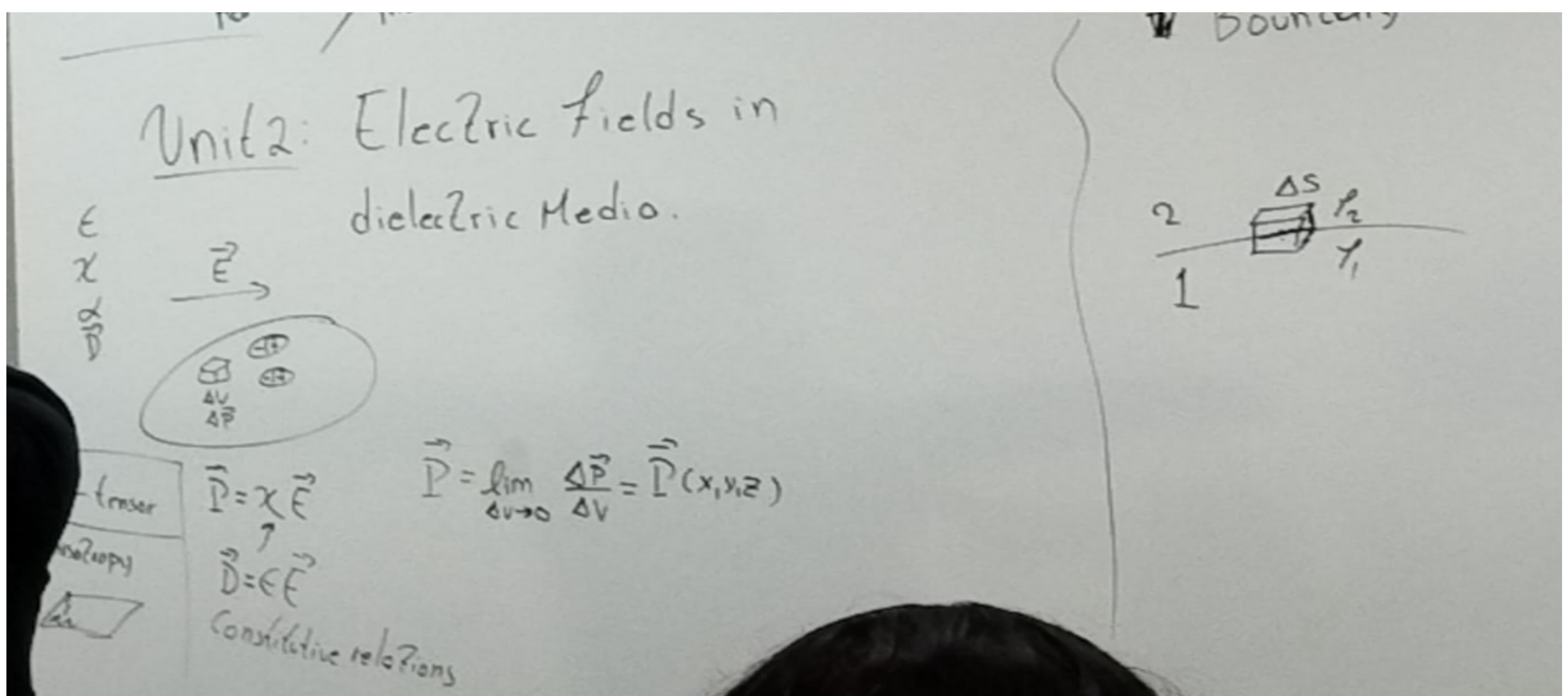
"Boundary conditions"

Densidad volumétrica de carga



$$\int_S \vec{D} \cdot \hat{n}\, da = q_f$$





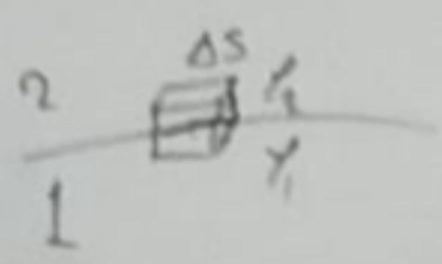
$$\int_S \vec{D} \cdot \hat{n}\, da = q_f$$

$\vdots$

$= \frac{1}{2} v (\rho_1 + \rho_2) +$

$\Delta S\, \sigma_f$

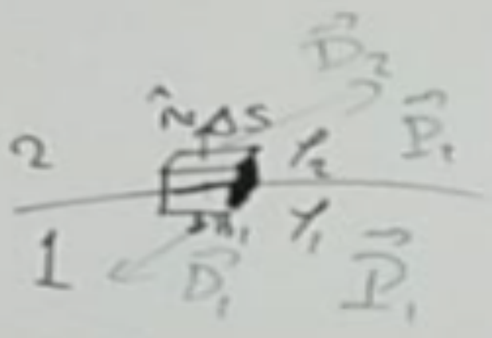
Boundary Conditions



$$\int_S \vec{D}\cdot\hat{n}\,da = q_f = \cancel{\tfrac{1}{2}v(\gamma_1+\gamma_2)} + \Delta s\,\sigma_f$$

$$E \propto \frac{1}{r^2} + \frac{1}{r^3} + \frac{1}{r^4}$$

Boundary Conditions



$$\int_S \vec{D}\cdot\hat{n}\,da = q_f = \cancel{\tfrac{1}{2}v(\gamma_1+\gamma_2)} + \Delta s\,\sigma_f$$

$$\vec{D} = \epsilon_0\vec{E} + \vec{P}$$

$$\vec{D}_2\cdot\hat{n}_2\,\Delta s + \vec{D}_1\cdot\hat{n}_1\,\Delta s = \Delta s\,\sigma_f$$

$$\hat{n}_2 = -\hat{n}_1$$

$$(\vec{D}_2 - \vec{D}_1)\cdot\hat{n}_2 = \sigma_f$$

$$\boxed{D_{2n} - D_{1n} = \sigma_f} \quad \checkmark\checkmark$$

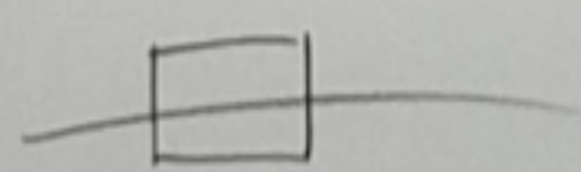
Considering that

$$U(\vec{r}) = \int_{\vec{r}_0}^{\vec{r}} \vec{E}\cdot d\vec{l}$$

$$\Rightarrow \oint_C \vec{E}\cdot d\vec{l} = 0$$

σ_f

Exercise:



E_{1t} E_{2t}

Considering that

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$$\Rightarrow \oint_C \vec{E}\cdot d\vec{l} = 0$$

σ_f

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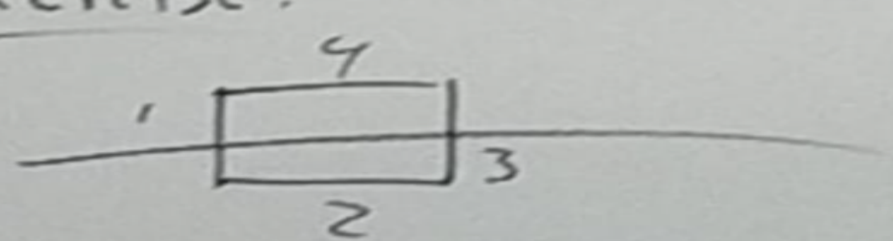

$$\boxed{E_{1t} = E_{2t}}$$

Considering that

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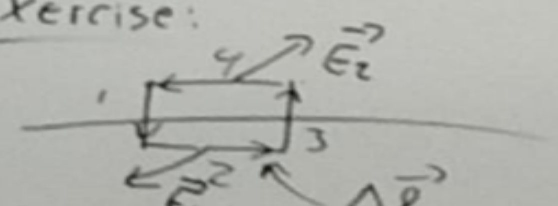
Exercise:



$$\boxed{E_{1t} = E_{2t}}$$

$$\vec{E}_1\cdot\Delta\vec{l} - \vec{E}_2\cdot\Delta\vec{l} = 0$$

$$(\vec{E}_1 - \vec{E}_2)\cdot\Delta\vec{l} = 0$$

Mañana veremos como resolver para dieléctricos:

$$\nabla^2 U = 0$$ • Laplace

$$\nabla^2 U = -\frac{\rho}{\varepsilon}$$ • Poisson